

DRY-RUN GUIDE | add_2ndlast_digit.java
Source: java\1st_belt\add_2ndlast_digit.java

CORE BEHAVIOR

Reads two integers, rejects either value below 10, otherwise extracts each tens digit with $(n/10)\%10$ and adds them. Negative inputs are also rejected by $n<10$.

REPRESENTATIVE INPUT / CALL

Input: '123423 123456'.

STEP TRACE

First tens digit= $(123423/10)\%10=2$; second= $(123456/10)\%10=5$; sum=7.

EXPECTED OUTPUT / RETURN

'7'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Challenge #11
003 |
004 | Imagine you have two numbers num1 and num2.
005 | Your task is to find the second-to-last digit of each number.
006 | If the numbers are valid, you'll add these two special digits together.
007 | But if either number is too small to have a second-to-last digit, you'll say it's an
    | "Invalid number".
008 |
009 | Steps:
010 | 1. Choose two numbers.
011 | 2. Check if these numbers are valid.
012 | 3. Find the second-to-last digit of each number.
013 | 4. Add these digits together.
014 | 5. If either number is too small (either of the number is less than 10 or a negative
    | number), print it's an "Invalid number".
015 |
016 | Example 1:
017 | Input:
018 | 123423
019 | 123456
020 |
021 | Output:
022 | 7
023 | */
024 |
025 | import java.io.*;
026 | import java.util.*;
027 |
028 | public class Solution {
029 |
030 |     public static void main(String[] args) {
031 |         Scanner sc = new Scanner(System.in);
032 |
033 |         // Read the two input numbers
034 |         int n1 = sc.nextInt();
035 |         int n2 = sc.nextInt();
```

```
036 |  
037 | // Check if either number is less than 10 or negative  
038 | if (n1 < 10 || n2 < 10) {  
039 |     System.out.println("Invalid number");  
040 | } else {  
041 |     // Find the second-to-last digit of each number  
042 |     int digit1 = (n1 / 10) % 10;  
043 |     int digit2 = (n2 / 10) % 10;  
044 |  
045 |     // Add the digits together and print the sum  
046 |     int sum = digit1 + digit2;  
047 |     System.out.println(sum);  
048 | }  
049 |  
050 |     sc.close();  
051 | }  
052 | }  
053 |
```